

Text Classification in NLP

CMPE346 — Group 2

Atakan Gül [Student 2] [Student 3]

June 5, 2026

Group task: Text Classification

Metric: Accuracy & Weighted F1

Student	Task
Atakan Gül	Media Bias Detection in News
[Student 2]	[Topic]
[Student 3]	[Topic]

BiLSTM

Bidirectional LSTM trained from scratch. Captures sequential context in both directions.

BERT

Transformer encoder pretrained via masked language modeling. Fine-tuned end-to-end. `bert-base-uncased`.

RoBERTa

Optimized BERT: removes next sentence prediction, uses dynamic masking and more data. `roberta-base`.

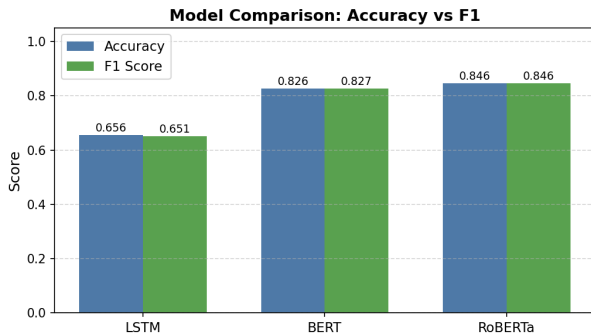
Task: Classify a news sentence as *biased* or *not biased*.

Dataset: BABE (Bias Annotations By Experts)

- ▶ 4,121 expert-annotated news sentences from diverse outlets
- ▶ 3,121 train / 1,000 test — binary labels

Model	Acc.	F1
BiLSTM	0.656	0.651
BERT	0.826	0.827
RoBERTa	0.846	0.846

- ▶ RoBERTa best: +19% over BiLSTM
- ▶ BERT close behind — both benefit from pretraining
- ▶ BiLSTM struggles on subtle linguistic bias



- ▶ Media bias is a nuanced task — framing and tone, not facts
- ▶ Pretraining is critical: transformers extract subtle linguistic patterns BiLSTM cannot
- ▶ RoBERTa's optimized pretraining gives it a clear edge over BERT on this task

Task: [Description]

Dataset: [Name]

Stats

/ test split

Model	Acc.	F1
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

[Figure or key observation]

inding 1

inding 2

Finding 1

Finding 2

Finding 3

Task: [Description]

Dataset: [Name]

Stats

/ test split

Model	Acc.	F1
BiLSTM	—	—
BERT	—	—
RoBERTa	—	—

[Figure or key observation]

inding 1

inding 2

Finding 1

Finding 2

Finding 3

All Results

Task	Model	Acc.	F1
3*Media Bias	BiLSTM	0.656	0.651
	BERT	0.826	0.827
	RoBERTa	0.846	0.846
3*[Student 2 Topic]	BiLSTM	—	—
	BERT	—	—
	RoBERTa	—	—
3*[Student 3 Topic]	BiLSTM	—	—
	BERT	—	—
	RoBERTa	—	—

Conclusion

- ▶ RoBERTa consistently outperforms across all three tasks and domains
- ▶ BERT and RoBERTa both adapt well regardless of classification topic
- ▶ BiLSTM fails to compete — pretraining is decisive on small labeled datasets
- ▶ Transformer-based models are robust general-purpose classifiers